Public Spending, Capital Accumulation, and Informality:

Why the Financing Instrument Matters in Colombia

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Abstract

This paper studies how a permanent increase in government purchases affects capital accumulation and labor informality. The distinction between spending and financing is central. I build an annual deterministic general-equilibrium model in which private capital is used only in formal production and workers choose between formal and informal employment. A smooth discrete-choice margin keeps both sectors active. The illustrative Colombian calibration matches 55.4% labor informality and government consumption equal to 14.7% of GDP. Government purchases then rise permanently by two percent of benchmark GDP. If the additional spending is financed by a lump-sum instrument, stationary capital and informality do not change, while private consumption falls 2.8 percent. If half of the increase is charged to the formal wage bill, capital falls 8.2 percent and informality rises 3.6 percentage points. If the entire increase is financed through the formal payroll, capital falls -22.1% and informality rises 9.8 percentage points. Whether public consumption enters utility additively changes welfare but not the allocation. Public spending must be nonseparable with private consumption, affect productivity, or modify an individual constraint to alter behavior independently of its financing. The quantitative results are mechanism exercises, not fiscal forecasts.

Keywords: government spending; informality; capital accumulation; payroll taxation; fiscal policy; Colombia. **JEL:** E22; E62; H30; H50; J46.

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1 Introduction

An increase in government spending withdraws resources from private uses, but that accounting statement does not determine its effect on capital or labor informality. The result depends on what the government purchases, whether households value it, and which tax base finances the additional expenditure. A payroll tax lowers the return to formal employment. A broad tax or a lump-sum instrument does not create the same formal–informal wedge. Debt can postpone the wedge but cannot eliminate the need for a fiscal adjustment after a permanent increase in spending.

This paper studies how a permanent increase in government purchases affects capital accumulation and labor informality in Colombia. The exercise compares three ways to finance the same increase in spending: a lump-sum instrument, an equal combination of lump-sum and payroll revenue, and a tax charged entirely to the formal wage bill. Every experiment starts from the same stationary equilibrium. The comparison therefore holds constant the size and timing of the spending increase and changes only its marginal source of financing.

The model is annual and deliberately parsimonious. A representative household chooses private consumption and saving. A unit mass of workers is allocated between formal and informal employment according to the difference between after-tax sectoral wages and heterogeneous sector preferences. Formal firms use capital and labor. Informal production is labor intensive. The aggregate capital stock follows from the household Euler equation and the economy's resource constraint.

The stationary calibration reproduces two Colombian aggregates. Labor informality is 55.4%, the proportion reported by Departamento Administrativo Nacional de Estadística (2025) for August–October 2025. General-government final consumption is 14.7% of GDP, close to the 2024 national accounts measure reported by World Bank (2026). The policy experiment raises government purchases by two percent of initial GDP. The remaining parameters are provisional and are selected from values commonly used in annual macroeconomic models. The exercise should consequently be read as a quantitative decomposition, not as an estimated response to a specific Colombian program.

The financing instrument determines the stationary response. Lump-sum financing leaves the formal employment wedge unchanged. Capital and informality therefore remain at their benchmark values, although private consumption falls 2.8 percent because government purchases absorb additional output. When half of the increase is financed from the formal wage bill, informality rises 3.6 percentage points, capital falls 8.2 percent, and output falls 3.8 percent. Full payroll financing increases informality by 9.8 percentage points and lowers capital by 22.1 percent. The spending increase is the same in every case; the differences come from the tax wedge and its effect on the formal labor input used together with capital.

The treatment of public consumption provides a second distinction. Preferences are

$$\sum_{t=0}^{\infty} \beta^t \left[u(C_t) + \chi_G \log G_t \right].$$

Because G_t is exogenous and additively separable, changing χ_G affects welfare but not consumption, saving, wages, or informality. This result is not an innocuous modeling detail. A paper that makes public consumption valuable by adding it to utility cannot claim a behavioral channel unless public and private consumption are nonseparable, the spending relaxes a household constraint, or government expenditure changes productivity.

The paper contributes a transparent decomposition rather than a new fiscal multiplier. Existing Colombian models contain richer nominal, fiscal, and labor-market blocks. The narrower structure used here isolates one question: how much of the response attributed to government spending is instead produced by the instrument used to finance it? The rest of the paper reviews the related literature, presents the model, describes the provisional calibration, reports stationary and transition results, and documents the numerical algorithm.

2 Related literature

The macroeconomic effect of government spending depends on household behavior, the persistence of the shock, and the fiscal instrument. In the neoclassical model of Baxter and King (1993), permanent government purchases affect private wealth, labor supply, and investment. Barro (1990) distinguishes unproductive government consumption from public services that enter production. Galí et al. (2007) shows that household heterogeneity and market structure can change the response of private consumption. The present paper sets aside short-run nominal rigidities and concentrates on capital accumulation and the formal–informal margin.

Informality makes the fiscal instrument part of the policy experiment. Levy (2008) emphasizes that payroll-financed benefits can tax formal employment while subsidizing activities outside the contributory system. La Porta and Shleifer (2014) and Ulyssea

(2018) show why informal firms cannot be treated only as formal firms that evade a tax. Productivity, technology, firm size, and selection differ across sectors. The current model retains only the labor-allocation margin and should be interpreted accordingly.

For Colombia, Mejía and Posada (2007) study the coexistence of formal and informal production when regulation can be evaded and enforcement is costly. Osorio-Copete (2016) uses a dynamic general-equilibrium model to study tax reform and informality. Rincón et al. (2017) construct a fiscal model that distinguishes government consumption from public investment and embeds a fiscal rule. Granda and Morales (2023) analyze tax policy, public enforcement spending, and labor-market rigidities. More recently, Herrera-Rojas et al. (2025) present a detailed fiscal model for Colombia with several revenue sources, expenditure categories, public debt, and a sovereign risk channel.

The present exercise is less detailed than those models. Its purpose is to make the financing decomposition visible. A permanent increase in G financed from a lump-sum source provides the resource benchmark. Charging an increasing share to formal payroll adds the endogenous-informality channel. This comparison would be harder to identify inside a model in which several tax rates, transfers, debt, and monetary policy adjust simultaneously.

3 Model

3.1 Household preferences and capital accumulation

Time is discrete and one period is one year. The representative household maximizes

$$U_0 = \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma} - 1}{1-\sigma} + \chi_G \log G_t \right], \qquad \beta \in (0,1), \quad \sigma > 0. \tag{1}$$

Private consumption is C_t. Government purchases are G_t. The public consumption term is additive and G_t is exogenous. It therefore does not enter the first-order condition for private saving:

$$\frac{C_{t+1}}{C_t} = \left[\beta(1 + r_{t+1})\right]^{1/\sigma}. \tag{2}$$

Changing χ_G changes welfare evaluated along a given allocation but does not change that allocation.

Private capital evolves according to the aggregate resource constraint:

$$K_{t+1} = (1-\delta)K_t + Y_t - C_t - G_t. \tag{3}$$

Government purchases are unproductive in the benchmark. They neither become public capital nor affect private total factor productivity.

3.2 Formal and informal production

Formal firms use private capital and formal labor:

$$Y_t^F = K_t^\alpha (L_t^F)^{1-\alpha}, \qquad \alpha \in (0,1). \tag{4}$$

The competitive formal wage and net return on capital are

$$w_t^F = (1-\alpha)\left(\frac{K_t}{L_t^F}\right)^\alpha, \tag{5}$$

$$r_t = \alpha\left(\frac{L_t^F}{K_t}\right)^{1-\alpha} - \delta. \tag{6}$$

Informal output uses labor only:

$$Y_t^I = A_I L_t^I, \qquad w_t^I = A_I. \tag{7}$$

The labor force has unit mass, so $L_t^I = 1 - L_t^F$. Total output is

$$Y_t = Y_t^F + Y_t^I. \tag{8}$$

Equation (7) is a benchmark, not a claim that informal firms literally use no equipment. It represents labor-intensive services and microenterprises with limited access to formal finance. The assumption also implies that a shift out of formal employment reduces the labor input used together with private capital. A robustness exercise should allow informal firms to use a smaller amount of capital and should introduce a financing wedge rather than assume frictionless mobility of capital across sectors.

3.3 Endogenous informality

Workers differ in their idiosyncratic valuation of the two sectors. Aggregating extreme-value preference shocks gives the formal employment share

$$L_t^F = \Lambda\left(\frac{\log[(1-\tau_t^F)w_t^F] - \log w_t^I - \kappa}{\mu}\right), \qquad \Lambda(x) = \frac{1}{1+e^{-x}}. \tag{9}$$

The parameter κ is the average nonpecuniary cost of formal employment. The scale μ controls heterogeneity. A positive μ prevents the economy from collapsing mechanically into a corner when after-tax wages change.

The formal payroll tax τ_t^F affects both the tax base and the allocation of labor. A higher tax lowers the net formal wage in (9). The resulting fall in L_t^F reduces the wage bill on which the tax is collected. The model therefore checks for an interior fiscal equilibrium rather than calculating the tax rate from a fixed base.

3.4 Government spending and fiscal closure

The government budget is balanced each period:

$$G_t = \tau_t^F w_t^F L_t^F + T_t, \tag{10}$$

where T_t denotes a lump-sum fiscal instrument. Let G_0 be benchmark spending and let $\Delta G_t = G_t - G_0$. Payroll revenue is required to satisfy

$$\tau_t^F w_t^F L_t^F = \lambda_0 G_0 + \lambda_\Delta \Delta G_t. \tag{11}$$

The remaining revenue is lump sum:

$$T_t = (1-\lambda_0)G_0 + (1-\lambda_\Delta)\Delta G_t. \tag{12}$$

All experiments use the same λ_0 and the same initial equilibrium. They differ in λ_Δ. The lump-sum experiment sets $\lambda_\Delta = 0$, the mixed experiment sets $\lambda_\Delta = 0.5$, and the payroll experiment sets $\lambda_\Delta = 1$. The lump-sum instrument is a theoretical benchmark. It is not proposed as a readily available Colombian tax.

3.5 Equilibrium

Given a path for government purchases and a marginal financing share, an equilibrium is a sequence

$$\{C_t, K_{t+1}, L_t^F, \tau_t^F, w_t^F, r_t, Y_t\}_{t=0}^{\infty}$$

that satisfies household optimality (2), capital accumulation (3), factor prices (5)–(6), endogenous formality (9), and the government budget (10)–(12).

In a stationary equilibrium,

$$r^\star = \frac{1}{\beta} - 1 \tag{13}$$

and

$$\frac{K^\star}{L^{F,\star}} = \left[\frac{\alpha}{\beta^{-1} - 1 + \delta}\right]^{1/(1-\alpha)}. \tag{14}$$

Equations (9), (11), and (14) jointly determine formal employment, the payroll tax, and capital.

4 Calibration and experiment

Table 1 reports the initial calibration. The model targets 55.4 percent national labor informality, measured by Departamento Administrativo Nacional de Estadística (2025) for August–October 2025. This target covers the national economy; it is higher than the corresponding rate for the thirteen largest metropolitan areas. The government-consumption target is 14.72 percent of GDP in 2024 (World Bank, 2026). It measures purchases of goods and services, not total public expenditure, transfers, or debt service.

The experiment is an unanticipated and permanent increase:

$$G_t = \begin{cases} G_0, & t = 0, \\ G_0 + 0.02Y_0, & t \geq 1. \end{cases} \tag{15}$$

The amount is fixed in units of benchmark output. It is not set equal to a constant share of contemporaneous GDP. Consequently, G_t/Y_t can rise by more than two percentage points if output falls.

Table 1: Initial calibration

Parameter	Symbol	Value	Status or target
Capital share	α	0.33	Provisional standard value
Annual depreciation	δ	0.06	Provisional standard value
Discount factor	β	0.96	Provisional standard value
Risk aversion	σ	2.00	Provisional standard value
Informal productivity	A_I	0.60	Normalization
Choice dispersion	μ	0.30	Provisional
Informal employment	L^I	0.554	DANE target
Government consumption / GDP	G/Y	0.1472	WDI target
Baseline payroll share	λ_0	0.50	Provisional closure
Permanent spending increase	$\Delta G/Y_0$	0.02	Policy experiment

Notes: The calibration is illustrative. Parameters labeled "provisional" have not been estimated for this model. The formality cost κ is selected to match the informality target conditional on the remaining parameters.

5 Results

5.1 Stationary effects

Table 2: Permanent public-spending increase: stationary effects

Financing of additional spending	ΔK (%)	ΔY (%)	ΔC (%)	Δ informality (p.p.)	$\Delta\tau_f$ (p.p.)
Lump-sum financing	0.00	0.00	-2.80	0.00	0.00
Mixed financing	-8.17	-3.83	-6.58	3.64	3.69
Payroll financing	-22.08	-10.35	-13.02	9.85	9.84

Notes: The increase in government purchases equals two percent of benchmark GDP. All scenarios share the same initial steady state. Lump-sum financing means that none of the additional spending is charged to the formal wage bill; mixed and payroll financing charge 50 and 100 percent, respectively.

Table 2 separates the resource effect of government purchases from the formal-tax effect. With lump-sum financing, the permanent increase in G reduces stationary private consumption by 2.8 percent. It does not change capital, output, or informality. The return condition (13) fixes the formal capital–labor ratio, while the formal employment wedge remains at its benchmark value. The additional government purchase is absorbed by lower private consumption.

Mixed financing changes the result because half of the additional expenditure is collected from the formal wage bill. The payroll tax rises 3.7 percentage points. Informality rises from 55.4 to 59.0 percent, an increase of 3.6 percentage points. Formal labor and capital are complements in production, so capital falls 8.2 percent. Output falls 3.8 percent and private consumption falls 6.6 percent.

Full payroll financing produces a larger contraction of the formal tax base. The payroll tax rises from 15.6 to 25.4 percent and informality reaches 65.2 percent. Capital falls 22.1 percent, output falls 10.4 percent, and private consumption falls 13.0 percent. These values are not estimates of a Colombian fiscal reform. They show the amplification produced by financing a permanent spending increase from a tax base that shrinks when its own rate rises.

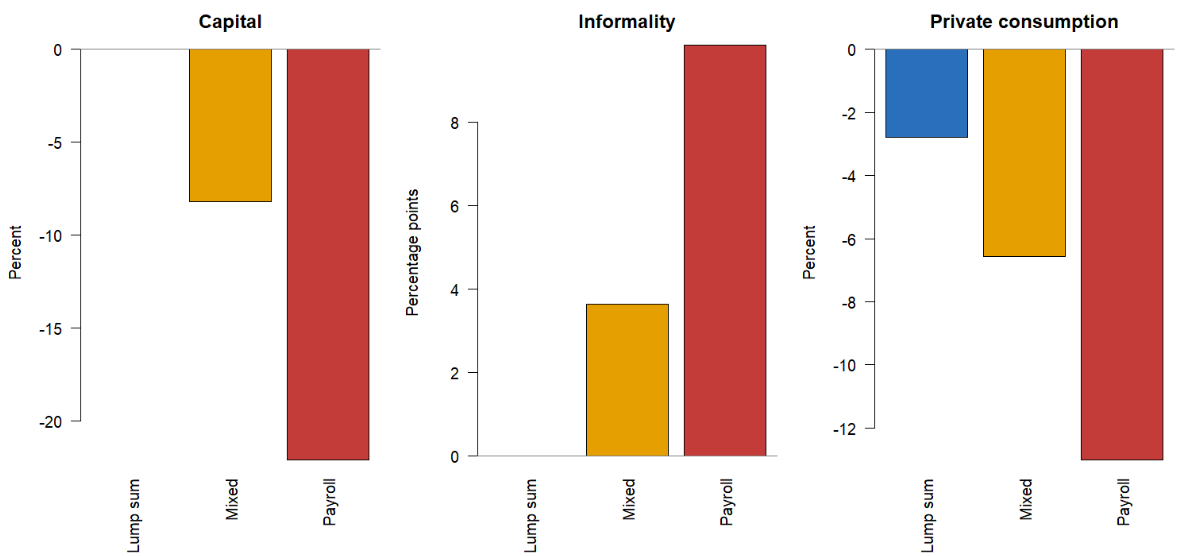

Figure 1: Stationary effects by marginal financing instrument

Notes: Changes are measured relative to the common benchmark. Government purchases rise permanently by two percent of initial GDP.

5.2 Transition

Figure 2 reports the perfect-foresight transition. Capital is predetermined when the spending increase is announced. Consumption and the fiscal instruments can adjust immediately. Under lump-sum financing, private consumption absorbs the spending increase on impact and capital remains at its initial stationary level. Under mixed and payroll financing, the higher formal tax reduces formal employment. Capital then declines gradually toward the lower stationary value.

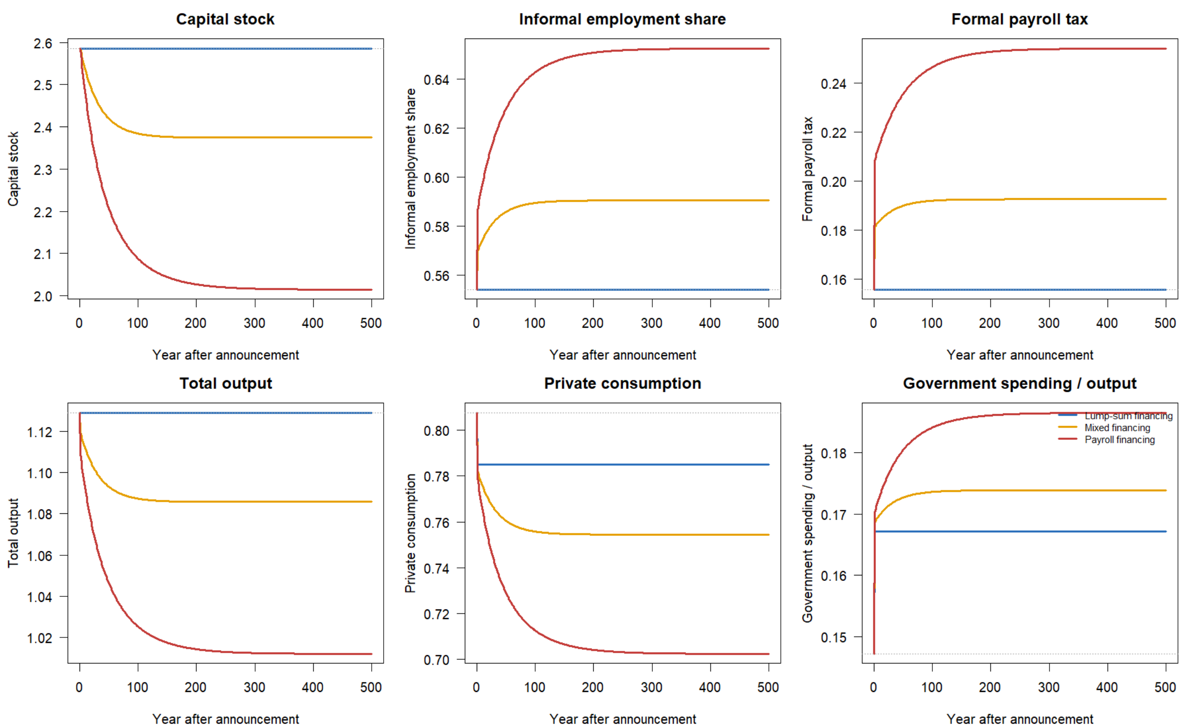

Figure 2: Transition after a permanent increase in government purchases

Notes: The gray dotted line is the common benchmark. The numerical solution uses a long terminal horizon because the stable root approaches the new stationary equilibrium slowly in the payroll-financed experiment. One period is one year.

The paths are monotone near the new steady state. They do not alternate between tax and informality regimes. The algorithm obtains the stable branch from the model's Euler and resource equations rather than applying a moving average or damping the plotted series. This distinction matters because a smooth figure is not evidence of an equilibrium transition unless the underlying residuals also converge.

5.3 Public consumption and welfare

The allocation is identical whether $\chi_G = 0$ or $\chi_G > 0$ in (1). The welfare assessment is not. With $\chi_G = 0$, a proportional increase of 2.9 percent in private consumption along the lump-sum-financed path would be required to restore benchmark welfare. The corresponding compensations are 4.8 percent under mixed financing and 7.4 percent under payroll financing. With $\chi_G = 0.15$, the value assigned to public consumption lowers those compensations to 1.3, 3.2, and 5.7 percent, respectively.

This comparison does not establish that the increase in G lowers Colombian welfare. The result depends on the provisional value of χ_G and on the assumption that government purchases are unproductive. Its narrower conclusion is that adding an exogenous and separable G_t to utility changes the welfare calculation without creating an additional effect on capital or informality.

6 Extensions and limitations

The first limitation is the fiscal closure. The government balances its budget every year. Public debt could transfer part of the adjustment across dates, but a permanent increase in spending still requires a higher stationary primary revenue. A debt extension should distinguish the transition from the long-run financing instrument and should impose a terminal debt rule.

The second limitation is the composition of spending. Public consumption is neither productive nor accumulated. A productive-spending extension could set

$$Y_t^F = A(G_t^P)K_t^\alpha (L_t^F)^{1-\alpha}, \qquad A'(G_t^P) > 0,$$

or introduce public capital with its own accumulation equation. The benefit of additional spending would then depend on its productivity, not only on the utility weight assigned to it.

The third limitation is the single final good. Formal and informal output are perfect substitutes in (8). A two-good extension should aggregate formal and informal consumption with a finite elasticity of substitution and should specify which good is used for investment and government purchases. This extension may change the size, but not the logical importance, of the financing channel.

The fourth limitation is the labor market. The logit allocation produces interior employment shares but does not contain unemployment, firm entry, search, wage rigidities, or enforcement. Those mechanisms are central in the richer Colombian literature. The current specification is useful only while the paper's purpose remains the decomposition of spending and financing.

7 Conclusions

A permanent increase in government purchases does not have a unique effect on capital accumulation or labor informality. The financing instrument is part of the policy. In the illustrative calibration, lump-sum financing lowers private consumption but leaves capital and informality unchanged. Charging half of the additional spending to the formal wage bill lowers capital 8.2 percent and raises informality 3.6 percentage points. Full payroll financing lowers capital 22.1 percent and raises informality 9.8 percentage points.

The model also clarifies what follows from placing government consumption in utility. When G_t is exogenous and additively separable, it changes welfare but not the allocation. To obtain an independent behavioral effect, public spending must interact with private consumption, affect productivity, or relax a constraint faced by households or firms.

The next quantitative step is not to add more adjustment parameters. It is to replace the lump-sum benchmark with observed Colombian tax instruments and a debt rule, distinguish public consumption from public investment, and estimate the formal-employment response to the relevant tax wedges. Those additions would permit the paper to move from a mechanism exercise to a Colombian policy counterfactual.

A Numerical algorithm

The code is written in R and uses only base packages. The stationary and transition calculations proceed as follows.

1. Given (K_t, G_t) and required payroll revenue, solve (9) jointly with (11). The algorithm searches all admissible roots and selects the root connected to the preceding allocation.

2. Compute formal and informal output, factor prices, tax revenue, and the government budget residual.

3. In a stationary equilibrium, impose (13) and solve the remaining scalar equation for formal employment. Compute consumption from

$$C^\star = Y^\star - G^\star - \delta K^\star.$$

4. For the transition, combine (2) and (3) into a two-dimensional map for (K_t, C_t). Capital at the announcement date is inherited from the benchmark.

5. Evaluate the Jacobian of that map at the final stationary equilibrium. The calibration has one stable and one unstable root. Select the eigenvector associated with the stable root.

6. Start close to the final equilibrium on the stable eigenvector and apply the inverse Euler and resource maps. Continue backward until the path reaches the inherited capital stock. A scalar root selects the terminal displacement that reproduces initial capital.

7. Reverse the resulting sequence and append the final stationary allocation after the terminal discrepancy is below the numerical tolerance. The reported tests verify the household Euler equation, the resource constraint, the government budget, the benchmark targets, and terminal convergence.

The stable-path method avoids the numerical instability of forward shooting. It does not smooth the endogenous variables after they are computed. The transition paths are generated by the model equations themselves.

B Alternative preference specifications

The additive specification in (1) can be replaced by a nonseparable private–public consumption aggregator:

$$X_t = \left[\omega^{1/\eta} C_t^{(\eta-1)/\eta} + (1-\omega)^{1/\eta} G_t^{(\eta-1)/\eta}\right]^{\eta/(\eta-1)}. \tag{16}$$

Utility would then be $u(X_t)$. Government consumption changes the marginal utility of private consumption and can affect saving during a transition. That model is economically different from assigning a positive value to χ_G in (1); it should be calibrated as a separate experiment.

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